Technical Documentation

KudiGuard — NAIC 2026, Problem Statement 2 (Voice-First Access)

Document 4 of the submission: architecture, setup and usage, sufficient for an independent party to run or integrate the build. Sources: `README.md`, `docs/ARCHITECTURE.md`, `docs/DEPLOY.md`, `.env.example`, `docs/ABUSE.md`, `docs/DPIA_VOICE.md`, `docs/CLAIMS.md`.

---

1. What it is

A money-safety assistant for Nigerians: send a voice note or a suspicious message in Hausa, Yoruba, Igbo, English or Pidgin; get red / orange / green with what to do and where to report. Channels: **WhatsApp** (voice notes, text, forwards, buttons), a **web voice page** at `/` (no install), and a **partner REST API** `/v1` (off in the pilot). Built on **N-ATLAS** (LLM) and the **NCAIR1** speech models, all run locally.

2. Architecture

```
channels                      core (no I/O assumptions)             adapters
--------                      -------------------------             --------
WhatsApp webhook --+          pipeline.Conversation                 llm/openai_compat -> N-ATLAS (llama.cpp, GPU)
Web page (/) ------+-> api/app -> (consent, commands, buttons,  ->  speech/natlas_asr -> NCAIR1 (CTranslate2,
CPU)
REST /v1 ----------+          limits, licence cap)                  knowledge/*       -> packs (files, partner
APIs)
                              engine.Engine (stateless check)       storage/sqlite    -> hashed, redacted log
                              safety/* (rules, verdict, guards)

 PC (Windows, NVIDIA GPU >= 8 GB)
 +-------------------------------------------------------------------------+
 | scripts/watchdog.ps1  -- starts / restarts --+--------------+-------------+ |
 |   llama-server :8081 (N-ATLaS Q4_K_M GGUF)   | KudiGuard    | cloudflared | |
 |   <-- OpenAI-compatible HTTP, localhost -->  | :8010        | HTTPS tunnel| |
 |                                              | FastAPI      |  -> public  | |
 |   var/kudiguard.db (SQLite)                  | + ASR on CPU |    URL      | |
 +-------------------------------------------------------------------------+
```

Engine flow (`src/kudiguard/engine.py`): greeting shortcut -> N-ATLAS ANALYSE (JSON) -> translate fallback -> deterministic verdict (`safety/decision.py`) -> knowledge search -> N-ATLAS COMPOSE -> output guard (retry once) -> fixed fallback. Time-budgeted: model steps that no longer fit are skipped. Yoruba/Igbo replies are composed in English and translated with fidelity and negation checks; Hausa/English are direct (`docs/ARCHITECTURE.md`).

Component	Module	Role
HTTP app	`api/app.py`, `channels/whatsapp_routes.py`	web page, WhatsApp webhook, /v1, health
Conversation	`pipeline.py`, `commands.py`	consent, STOP/DELETE/language commands, buttons, ordering, rate limits, never-silent failures
Admission	`admission.py`	N-ATLaS licence cap (1,000 active users / rolling 30 days) across all channels
Engine	`engine.py`, `prompts.py`, `reply_translation.py`	the check: analyse, verdict, search, compose, translate
LLM client	`llm/base.py`, `llm/openai_compat.py`, `llm/sampling.py`	N-ATLAS calls with purposes, budgets, temperature 0 / seed 42
Speech	`speech/audio.py`, `speech/natlas_asr.py`, `speech/service.py`	decode, trim, voiced-length check, NCAIR1 ASR, asr_terms

Component	Module	Role
Safety	`safety/scam_rules.py`, `safety/decision.py`, `safety/output_guard.py`, `safety/guard_classifier.py`, `safety/grounding.py`	rules, verdict policy, reply guards
Knowledge	`knowledge/*`, `data/packs/*`	fact-checked packs, hybrid (BM25 + bge-small) search
Languages	`data/languages/{en,ha,yo,ig}.yaml`	lexicons, ASR model names, asr_terms, UI strings, abuse phrases
Storage	`storage/sqlite.py`, `privacy.py`, `text.py`	hashed ids, redacted text, retention purge

3. Setup (Windows, NVIDIA GPU >= 8 GB)

From `README.md` and `docs/DEPLOY.md`. Tools and models are not in git:

- `tools\llama\`: llama.cpp release b11476 win-cuda-12.4 (`llama-server.exe` + CUDA DLLs).
- `models\N-ATLaS-GGUF-Q4_K_M.gguf` from Hugging Face `tosinamuda/N-ATLaS-GGUF`, SHA256 `1498091db609d8db8bf4637603fe8c705af17476a465b554c9c44e0ae2e69881`.
- `tools\ffmpeg\` (any LGPL build, or ffmpeg on PATH); `tools\cloudflared\cloudflared.exe` 2026.9.1.

```
python -m venv .venv; .venv\Scripts\pip install -e ".[dev,semantic,asr]"
copy .env.example .env                        # fill in; KUDIGUARD_HASH_SECRET is required
.venv\Scripts\python scripts\prepare_asr.py   # one-time: NCAIR1 speech models -> CTranslate2 int8 (HF_TOKEN)
.venv\Scripts\python -c "from kudiguard.knowledge.embeddings import FastEmbedder as F;
F(cache_dir='models/fastembed').embed_documents(['warm up'])"
.\scripts\watchdog.ps1                        # N-ATLAS (8081) + KudiGuard (8010) + HTTPS tunnel; restarts what
stops or hangs
```

Then (WhatsApp): Meta -> WhatsApp -> Configuration -> Webhook URL `https://<tunnel-host>/webhooks/whatsapp`, verify token from `.env`, subscribe to **messages**. Check `https://<tunnel-host>/readyz` -> `{"status":"ok", ...}`. `scripts/setup_named_tunnel.ps1` + `watchdog.ps1 -Named` give a stable URL. N-ATLAS alone: `scripts/run_llm.ps1`.

Hardware used: RTX 4070 Laptop GPU 8 GB (N-ATLAS uses 7.5 GB; speech runs on CPU), `docs/SPIKES.md` S1.

Cloud (later): `docker build -t kudiguard .`, same env vars, volume for `/app/var`; any GPU host running llama.cpp or vLLM with the same GGUF via `KUDIGUARD_LLM_URL` (`docs/DEPLOY.md`).

3.1 `.env` keys (from `.env.example`; values omitted)

Group	Keys
Privacy	`KUDIGUARD_HASH_SECRET` (required), `KUDIGUARD_STORAGE_LOCATION`, `KUDIGUARD_RETENTION_DAYS`, `KUDIGUARD_PATTERN_RETENTION_DAYS`, `KUDIGUARD_ALLOW_LIVE_PACKS`, `KUDIGUARD_VOICE_DONATION`, `KUDIGUARD_VOICE_NO_REASSURANCE`, `KUDIGUARD_VOICE_KEY`, `KUDIGUARD_VOICE_RETENTION_DAYS`, `KUDIGUARD_VOICE_MAX_PER_DAY`, `KUDIGUARD_VOICE_MAX_GB`, `KUDIGUARD_VOICE_DIR`
WhatsApp Cloud API	`WHATSAPP_PHONE_NUMBER_ID`, `WHATSAPP_ACCESS_TOKEN`, `WHATSAPP_APP_SECRET`, `WHATSAPP_VERIFY_TOKEN`, `WHATSAPP_API_VERSION`, `WHATSAPP_TEST_RECIPIENTS`
Models	`HF_TOKEN`, `HF_HOME`, `KUDIGUARD_LLM_URL`, `KUDIGUARD_LLM_API_KEY`, `KUDIGUARD_MODEL_CACHE`, `KUDIGUARD_EMBEDDINGS_CACHE`, `KUDIGUARD_SPEECH`, `KUDIGUARD_PRELOAD_LANGUAGES`, `KUDIGUARD_FFMPEG`, `KUDIGUARD_GUARD_CLASSIFIER_PATH`
Reply checks	`KUDIGUARD_GROUNDING_MODE` / `_THRESHOLD` / `_BUDGET_MS`, `KUDIGUARD_GUARD_CLASSIFIER_MODE` / `_PIDGIN_MODE` / `_THRESHOLD` / `_BUDGET_MS`

Group	Keys
Advice quality	KUDIGUARD_COMPOSE_STRATEGY, KUDIGUARD_KNOWLEDGE_MIN_RELEVANCE_DENSE / _BM25, KUDIGUARD_COMPOSE_EXAMPLES, KUDIGUARD_COMPOSE_EXAMPLE_MIN_RELEVANCE_DENSE / _BM25, KUDIGUARD_COMPOSE_EXAMPLE_COPY_MAX, KUDIGUARD_NOTICE_PREFIX, KUDIGUARD_NOTICE_MIN_RELEVANCE_DENSE / _BM25, KUDIGUARD_PIDGIN_REPLIES
Data, channels, limits	KUDIGUARD_DATA_DIR, KUDIGUARD_DB_PATH, KUDIGUARD_PACKS, KUDIGUARD_ALLOW_SAMPLE_PACKS, KUDIGUARD_WHATSAPP_ENABLED, KUDIGUARD_PUBLIC_URL, KUDIGUARD_DEFAULT_LANGUAGE, KUDIGUARD_PER_USER_PER_HOUR, KUDIGUARD_TURN_DEADLINE_S, KUDIGUARD_ACTIVE_USER_CAP, KUDIGUARD_WEB_SESSION_CAP, KUDIGUARD_API_RATE_PER_MINUTE, KUDIGUARD_WEB_RATE_PER_MINUTE, KUDIGUARD_TRUST_PROXY_HEADERS
Partner API	KUDIGUARD_PARTNER_API, KUDIGUARD_API_KEYS
Malicious use (off/shadow/enforce)	KUDIGUARD_ABUSE_CRISIS_MODE, KUDIGUARD_CRISIS_SEMANTIC_MODE / _THRESHOLD, KUDIGUARD_ABUSE_HARASSMENT_MODE, KUDIGUARD_ABUSE_SCAM_TOOL_MODE, KUDIGUARD_ABUSE_INJECTION_MODE, KUDIGUARD_ABUSE_INJECTION_SUSPECT_MODE, KUDIGUARD_REPLY_SCREEN_MODE, KUDIGUARD_REPLY_ECHO_WORDS, KUDIGUARD_ASR_HALLUCINATION_MODE, KUDIGUARD_PACK_SCREEN_MODE, KUDIGUARD_PACK_SCREEN_CORE_MODE, KUDIGUARD_LEARNING_GATE_MODE, KUDIGUARD_PATTERN_MAX_PER_USER_PER_DAY, KUDIGUARD_PATTERN_MAX_SIMILAR_PER_WEEK, KUDIGUARD_NEW_USER_MODE, KUDIGUARD_NEW_USERS_PER_HOUR / _PER_DAY, KUDIGUARD_FLOOD_MODE, KUDIGUARD_MAX_QUEUED_PER_USER, KUDIGUARD_RATE_NOTICE_INTERVAL_S, KUDIGUARD_VOICE_NOTES_PER_USER_PER_HOUR
Public pages	KUDIGUARD_OPERATOR_NAME, KUDIGUARD_PRIVACY_CONTACT

3.2 Settings added or decided on 12 Oct 2026 (live pilot)

Setting	Default	Effect
KUDIGUARD_PRELOAD_LANGUAGES	ha,en (config.py)	speech models loaded at startup, so the first voice note does not wait for a model load
KUDIGUARD_VOICE_NO_REASSURANCE	true	a non-English voice note with no flags gets "I may not have heard every word ... please type your message" instead of the green "no red flags" line; presentation only
KUDIGUARD_FOCUSED_RISK_CHECK	off (experiment branch only, not merged)	extra N-ATLAS risk call on ha/yo/ig originals; measured on dev2 and rejected
MIN_VOICED_SECONDS (code constant)	0.5 s	shorter voice notes are refused as "too short"

4. Usage

4.1 WhatsApp

Message the number; accept the welcome (consent); choose a language; send a voice note, text or forward. Commands: STOP (stop), DELETE (erase your data), language change; buttons "What should I do?", reporting, "Warn my family". Meta's free test number reaches at most 5 verified recipients (docs/DEPLOY.md).

4.2 Web voice page

Open https://<host>/, pick a language, record or type. The page posts to POST /web/check; GET /privacy shows the privacy notice with the configured retention values; POST /web/donation/withdraw withdraws a voice donation.

4.3 Partner REST API (/v1, off in the pilot)

Mounted only when KUDIGUARD_PARTNER_API=true with KUDIGUARD_API_KEYS (comma-separated name:key pairs).

Method	Path	Purpose
POST	`/v1/check`	check a message, JSON `{"text": "...", "language": "ha"}` (`language`: ha, yo, ig, en; default en)
POST	`/v1/check/audio`	check a voice note (multipart upload)
POST	`/v1/ask`	money-safety question, JSON `{"question": "..."}`
GET	`/v1/languages`	supported languages
GET	`/v1/topics`	knowledge topics the answers are grounded on

Headers on every call: **`X-API-Key`** and **`X-KudiGuard-End-User`**: a stable pseudonymous id for the partner's customer, 8–137 characters (letters, digits, `. _ : -`), e.g. a SHA-256 hex digest or UUID, optionally `c-<sha256>`. Phone- or email-shaped ids are refused (400). Each end user counts toward the shared licence cap; beyond it: 503 `at_capacity`, or 503 `busy_new_users` (`Retry-After: 3600`) from the new-user gate. Health: `GET /healthz`, `GET /readyz`. OpenAPI schema from FastAPI.

5. Testing and quality

Check	Command	Latest result
Unit + integration tests, coverage	`.venv\Scripts\python -m pytest --cov=kudiguard --cov-branch`	5,533 passed, 1 skipped, 12 xfailed; **100% line + branch** (9,455 statements, 2,258 branches) (latest full run, master 17a4e13/b6cfd4e, 12 Oct 2026)
Lint / format	`ruff check`, `ruff format --check`	ruff clean (same run)
Types	`mypy`	clean (same run)
Live model tests	`python -m pytest -m live`	need N-ATLAS and the speech models; not counted above
Frozen benchmark	`PYTHONIOENCODING=utf-8 python scripts/eval_e2e.py --summary data/eval/results/e2e-live.jsonl`	T v1, document 3
Mutation testing	`python scripts/mutation.py text legacy`	ten safety-critical modules 93.1% (round 3), `docs/CLAIMS.md` C9

Tests use fake models; coverage shows lines ran, not that behaviour is right (`docs/CLAIMS.md` C8).

6. Privacy and security summary

- **Identifiers**: phone numbers stored only as keyed hashes (`privacy.py`, `KUDIGUARD_HASH_SECRET`); logs never carry the WhatsApp message id (`docs/CLAIMS.md` P3).
- **Message text**: most numbers and emails masked before storage (`text.redact`); names and addresses are kept; known residual leak classes listed in `docs/CLAIMS.md` C13.
- **Voice**: voice notes deleted after transcription unless the user opts in to donation (off until the DPIA is signed off; `docs/DPIA_VOICE.md`); donated audio is encrypted (`KUDIGUARD_VOICE_KEY`), kept at most 12 months.
- **Retention**: messages ~90 days, scam examples under 2 years, purge every 6 hours while the server runs; users send `DELETE` or `STOP`.
- **Local models**: N-ATLAS, NCAIR1 and the embedding model run on our server; messages are not sent to outside AI companies (`docs/CLAIMS.md` P9). Pilot data is stored outside Nigeria and users are told so (`KUDIGUARD_STORAGE_LOCATION=outside-nigeria`).
- **Webhook security**: every WhatsApp webhook is verified with `X-Hub-Signature-256` (`WHATSAPP_APP_SECRET`).
- **Malicious use** (`docs/ABUSE.md`): crisis messages get help numbers first; abuse, scam-writing requests and prompt injections are screened; replies are screened for bank persona claims and prompt leaks; partner

packs are screened at load; rate limits per user, per IP and for new users. Each defence is `off | shadow | enforce`.

- **Safety of advice**: the verdict comes from rules, not the model; KudiGuard never says a message is safe; the model never writes contacts; output guard on every reply (`docs/ARCHITECTURE.md`).
- **Licence**: N-ATLaS cap of 1,000 active end-users per rolling 30 days enforced in code (`admission.py`).
- **Limits**: Hausa/Yoruba/Igbo lexicons, guard patterns and texts are DRAFT pending native review; one server process (SQLite); one GPU serves ~10–12 checks a minute (`docs/CLAIMS.md` C11).